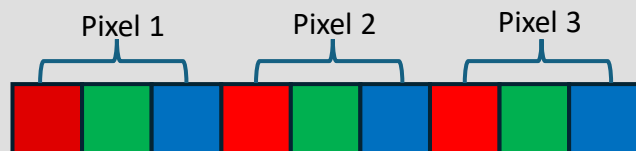


Data Layout Transformations

- N: Number of images in the batch.
- C: Number of Channels per image (e.g., 3 for Red, Green, Blue).
- H: Height of the image in pixels.
- W: Width of the image in pixels.

NHWC



- **Order:** Batch, Height, Width, Channels
- **What it means:** For each pixel, all its color/feature values are stored next to each other.
- **Typical:** TPUs/mobile NPUs; kernels that combine channels at a pixel
- Order in the above image: pixel 1(red)-> pixel 1 (green), pixel 1 (blue), pixel 2(red), pixel 2 (green), pixel 2 (blue), pixel 3(red), pixel 3 (green), pixel 3 (blue)

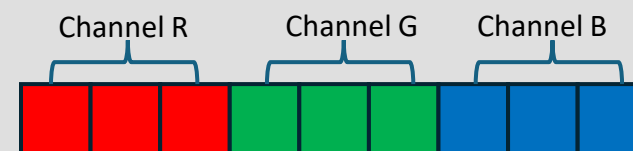
Reorder/transpose axes



Example permutation: [0, 3, 1, 2]
turns NHWC into NCHW

*No data changes—just the order
we store it.*

NCHW



- **Order:** Batch, Channels, Height, Width
- **What it means:** For each channel, all the pixels in the image are stored next to each other.
- **Good when:** You sweep over images one channel at a time.
- **Typical:** Convolution on many GPUs

For large matrix multiplies, we often go beyond NHWC/NCHW and use packed tile layouts that match the GEMM kernel

GEMM stands for General Matrix Multiplication. It's the mathematical name for the core operation that powers deep learning and scientific computing: multiplying two large matrices together.